

Read the following and answer the questions as follows: Question Numbers: (1 to 5)

The following table shows the cumulative percentage of the total number of students appearing in the Hindi and Punjabi examination and the cumulative number of students appearing in the Hindi examination and scoring marks in five different categories, namely, 80 and above; 70 and above; 60 and above; 50 and above; and 40 and above. Based on the data in the table, answer the questions that follow.

Marks-wise distribution of students

Marks (Out of 100)	Cumulative percentage of the total number of students who appeared in Hindi and Punjabi	Cumulative number of students who appeared in Hindi
80 and above	15%	108
70 and above	35%	264
60 and above	65%	424
50 and above	90%	554
40 and above	100%	624

Note:

- (1) Total number of students who attended the exam is 1200.
- (2) No student failed in the examination.
- (3) Each student appeared in only one exam, that is, either in Hindi or in Punjabi exam.

1. What is the sum of the number of students who scored 70 or more than 70 but less than 80 marks in Hindi and the number of students who scored 80 or more than 80 marks in Punjabi?

- (1) 330
- (2) 228
- (3) 272
- (4) 284

Answer Key: (2)

Solution:

Students scoring 70–80 in Hindi

Hindi students scoring:

- 70 and above = 264

- 80 and above = 108

So,
 $264 - 108 = 156$

Students scoring 80 and above in Punjabi
 15% of total students scored 80 and above.
 15% of 1200:
 $15/100 \times 1200 = 180$
 Out of these, Hindi students = 108
 So, Punjabi students:
 $180 - 108 = 72$

Required sum
 $156 + 72 = 228$
 Hence, the correct answer is **228**.
Hence, option (2) is correct.

- 2.** What is the ratio of the number of students who scored 50 or more than 50 marks but less than 60 marks in Hindi to the number of students who scored 40 or more than 40 marks but less than 50 marks in Punjabi?
- (1) 12:11
 - (2) 8:7
 - (3) 3:1
 - (4) 13 :5

Answer Key: (4)

Solution:

Hindi students scoring 50–60

Hindi students:

- 50 and above = 554
- 60 and above = 424

So,

$$554 - 424 = 130$$

Punjabi students scoring 40–50

Total students scoring:

- 40 and above = 1200
- 50 and above = 90% of 1200

$$90/100 \times 1200 = 1080$$

So, students scoring 40–50 overall:

$$1200 - 1080 = 120$$

Hindi students scoring 40–50:

$$624 - 554 = 70$$

So, Punjabi students scoring 40–50:

$$120 - 70 = 50$$

Required ratio

130:50

Divide by 10:

13:5

Hence, the correct answer is **13: 5**.

Hence, option (4) is correct.

3. The number of students who scored 80 and above marks in Hindi is _____% less than the number of students who scored 60 or more than 60 marks but less than 70 marks in Punjabi.

(1) 46

(2) 54

(3) 44

(4) 52

Answer Key: (1)

Solution:

Hindi students scoring 80 and above

Given directly: 108

Punjabi students scoring 60–70

Total students scoring:

- 60 and above = 65% of 1200

$$65/100 \times 1200 = 780$$

- 70 and above = 35% of 1200

$$35/100 \times 1200 = 420$$

So total students scoring 60–70:

$$780 - 420 = 360$$

Hindi students scoring 60–70:

$$424 - 264 = 160$$

So Punjabi students scoring 60–70:

$$360 - 160 = 200$$

Percentage less

Formula:

Difference / Larger value $\times 100$

$$200 - 108 = 92$$

$$92/200 \times 100 = 46\%$$

Hence, the number of Hindi students scoring 80 and above is **46% less.**

Hence, option (1) is correct.

4. What is the difference between the number of students who scored 70 or more than 70 marks but less than 80 marks in Punjabi and the number of students who scored 40 or more than 40 marks but less than 50 marks in Hindi?

(1) 20

(2) 16

(3) 14

(4) 30

Answer Key: (3)

Solution:

Punjabi students scoring 70–80

Total students scoring:

- 70 and above = 35% of 1200, $35/100 \times 1200 = 420$
- 80 and above = 15% of 1200, $15/100 \times 1200 = 180$

So total students scoring 70–80:

$$420 - 180 = 240$$

Hindi students scoring 70–80:

$$264 - 108 = 156$$

So Punjabi students scoring 70–80:

$$240 - 156 = 84$$

Hindi students scoring 40–50

$$624 - 554 = 70$$

Difference

$$84 - 70 = 14$$

Hence, the correct answer is **14.**

Hence, option (3) is correct.

5. What is the average of the number of students who scored 50 or more than 50 marks but less than 60 marks in Punjabi and the number of students who scored 50 or more than 50 marks but less than 60 marks in Hindi?

- (1) 180
- (2) 150
- (3) 200
- (4) 140

Answer Key: (2)

Solution:

Hindi students scoring 50–60

$$554 - 424 = 130$$

Punjabi students scoring 50–60

Total students scoring:

- 50 and above = 1080
- 60 and above = 780

So total students scoring 50–60:

$$1080 - 780 = 300$$

Punjabi students scoring 50–60:

$$300 - 130 = 170$$

Average

$$130 + 170 = 300$$

$$300/2 = 150$$

Hence, the correct answer is **150**.

Hence, option (2) is correct.